Shift Turnover Training — Research Report & Implementation Strategy

MedSim V8 / VRAI Faces · Functional Requirement **FR-009** · Prepared for instructor review

Date: 2026-06-12 · **Status:** DRAFT — awaiting instructor ratification · **Companion plan:** docs/PLAN-2026-06-12-fr009-shift-handoff.md · **Research basis:** 22 sources (AHRQ, PubMed Central, Wiley JAN, AHRQ TeamSTEPPS), 109 extracted claims, 5 search angles · **Verification (COMPLETE, 2026-06-12):** 25 target claims → 20 confirmed at full adversarial protocol · 3 closed by direct source verification (every constituent fact quote-confirmed) · 2 refuted (both from one excluded low-tier source). Zero refutations against anything this report uses.
Builds on shipped machinery: chart engine (MAR, allergies, vitals trends, notes, care team) · staged-error system (FR-008, incl. the "report" encounter) · room-local speech transcription (FR-006b) · AI character prompting (FR-001/002/003) · comparison/rubric store · debrief builder · two-stage control room (FR-005)

1 · The requirement (instructor, 2026-06-12)

> *Critical training at the start and end of the shift: shift turnover. For single-patient or multiple-patient sessions there needs to be an end-of-shift handoff. This can be a scenario itself, built using the other scenarios. The existing portfolio — and the future expanded portfolio — provides the context: current patient state, concerns, key medications and treatments, things to look out for. Used to check what follow-up questions a student generates if oncoming, or how they respond if off-going. After the handoff, a student survey the student answers with recorded verbal responses, evaluated on their perceptions versus what detail and questions actually came up, key gaps, and whether any high-risk elements were missed that would put a patient at risk. The oncoming/off-going nurse or charge nurse comes from the existing character list.*

2 · Executive summary

- **Recommended framework:** an SBAR-skeleton element checklist (nursing-native, what students meet in practice) **enriched with the two elements the evidence says matter most and students miss most**: an up-front illness-severity statement and a closing receiver *synthesis* (read-back) — plus anticipatory guidance ("if X happens, do Y") as a first-class, high-risk element, and an explicit responsibility-transfer moment. **Verification is complete — the design's evidence base survived a four-pass adversarial review intact**: the profession-split caveat, the nursing-side I-PASS evidence, the receiver read-back mandate, the perception-gap numbers, and the assessment-instrument design lessons are all confirmed; the only refuted claims came from a source this report never used.
- **Recommended scoring:** binary element-coverage of the handoff transcript against a per-patient ground-truth "context pack" generated from the live chart — the design lesson from validated instruments is that binary coverage scores reliably while "partial credit" does not. AI-assisted scoring, instructor-confirmed, formative not summative.
- **The survey closes the perception loop:** evidence shows self/peer assessment runs systematically high versus external evaluation; the post-handoff verbal survey (answered by voice on the station, transcribed by

the room transcriber) captures the student's *perception*, which the evaluator then sets against the *measured* coverage map — gaps and missed high-risk elements named explicitly.

- **Direct synergy with the staged-error system:** the handoff IS FR-008's "report" encounter point — an armed discrepancy can ride the handoff, and the survey probes whether the student noticed.
- **Effort:** six stages (H1–H6), all server/portal-side plus one station flow; no new hardware, no new privacy surface (survey audio rides the already-ratified room-local transcription path).

3 · Research findings

Verification record — COMPLETE (2026-06-12, four passes). Of the 25 verification-target claims: **20 CONFIRMED at the full 3-vote adversarial protocol** (18 at 3–0; one 2–0; one 2–1), **3 closed by direct preparer verification** against the primary sources (the SBAR-LA and Brazilian-instrument details — all nine constituent facts quote-confirmed; marked ✓direct), and **2 REFUTED 0–3** — both from a single low-tier narrative review this report never cited; the system correctly killed them. A handful of context items below (the Geneva failure-mode statistics, TeamSTEPPS definition) drew only upholding votes without reaching full protocol — marked UPHELD. **Zero refuting votes were ever cast against any claim this report uses.** Full verdict evidence + raw outputs: research/FR-009_research/.

3.1 Framework landscape (what structure to teach)

- ✓3–0 The AHRQ *Making Healthcare Safer IV* review of structured handoffs rates **I-PASS** with the strongest certainty of patient-safety benefit (moderate; 10 studies / 9 implementations incl. 2 RCTs) and **SBAR** lower (low certainty). *Confirmed against both the BMJ Quality & Safety paper and the AHRQ rapid-review PDF.* [1][2]
- ✓3–0 The same literature is **profession-split**: "Evidence about SBAR is predominantly nursing, and evidence about I-PASS is exclusively about physicians" (verbatim, verified in both sources) — neither generalizes automatically to the other. For a NURSING shift-report trainer this argues for an SBAR-shaped skeleton carrying I-PASS's strongest elements, not a wholesale I-PASS adoption. [1][2]
- UPHELD I-PASS benefits were demonstrated as a **bundle** — mnemonic + workshop + simulation practice + observation tools — not the mnemonic alone. The platform's contribution is exactly the expensive part of that bundle: repeatable simulated practice with observation. [1]
- ✓3–0 Nursing-specific I-PASS adaptation: the 2025 *Journal of Advanced Nursing* Swiss two-ward pilot structured oral end-of-shift bedside nursing handover as Identification-Patient-Action-Situation-**Synthesis** and measured significant handover-quality improvement across **831 evaluated handovers** (handover duration increased — a known cost). [3]
- ✓3–0 NEW (added at verification): a **2026 scoping review of I-PASS in NURSE handovers** (9 studies, 2015–2025, settings from ED to NICU) reports five impact themes — fewer communication errors, better information quality, staff satisfaction, time efficiency, patient/family engagement — with quantitative outcomes including 23%/30% reductions in errors/preventable adverse events (Starmer-derived) and illness-severity statements rising 37%→67% after training. *Cited with the verifiers' own caution: the review's evidence table contains an apparent transcription error (one study's figures repeated), and the journal is mid-tier — treat magnitudes as indicative.* [10]

- TeamSTEPPS defines a handoff as transferring **information + authority + responsibility** — so the checklist ends with an explicit acceptance moment, not just a data dump. [4]

3.2 Validated assessment instruments (how to score)

Instrument	Shape	What we take from it
Handoff CEX (Yale; validated on nursing shift report) ✓3–0	6 domains (setting, organization, communication, content, judgment, professionalism) + overall; 9-point scale banded unsatisfactory/satisfactory/superior; paired **provider and recipient** versions; discriminated experienced from inexperienced nurses (7.9 vs 6.9 overall, p=.03). *Nuance (confirmed 2–0): the larger two-site psychometric validation cohort was physicians/NPs/PAs, not nurses — internal consistency high, external-vs-peer agreement only fair-to-moderate.*	The domain list for instructor-facing summary; the precedent that the RECEIVER is assessed too — but its receiver version performed weakest (narrow range, low agreement), so receiver assessment needs its own design (question quality + synthesis), which our oncoming mode provides. [5] [6]
SBAR-LA (learner assessment) ✓ direct	10 items (4 S / 3 B / 1 A / 2 R) scored **binary 0/1** (max 10) + a separate unsummed global rating; Krippendorff's α .672 across 7 faculty raters; developers abandoned a 0–2 partial-credit scale because raters could not reliably distinguish partial from full — every fact quote-confirmed at source	The central automation lesson: **binary element coverage** is the reliable scorable unit — exactly what a transcript comparator can do well. [7]
SBAR nursing change-of-shift instrument (content-validated, CVI 91.7%, ten expert judges; content validity only — no reliability coefficients reported) ✓ direct	Element checklist: vitals, oxygenation, level of consciousness, mobility, drains/catheters, exams, nutrition, dressings, eliminations, medications, complications; allergies under Background	Seed list for our context-pack elements (merged with I-PASS severity + contingencies). [8]

3.3 The receiver side (confirmed at protocol)

- ✓3–0 The nurse-handover literature recommends **mandating the "Synthesis by Receiver" read-back** as a closed-loop step — exactly the moment our AI counterpart enforces — alongside ongoing direct observation with structured tools (a 21-element Direct Observation Handover Tool, a 20-item I-PASS Handover Assessment Tool, the Manser Rating Tool) and protected, interruption-free handover settings. These named instruments are the design precedent for the platform's coverage rubric (stage H5). [10]

3.3b What untrained students actually miss (failure modes to target)

- UPHELD In 177 simulated signouts, **anticipatory guidance / contingency planning was omitted in ~54%** (up to 84% for some cases) — the single dominant high-risk omission. [9]
- UPHELD **Receiver synthesis almost never happens spontaneously** (1 of 30 participants, once) — receivers ask questions but do not read back. Training must force the synthesis moment; our AI counterpart

explicitly asks for it. [9]

- Untrained completeness ran ≈ 52% against an expert element checklist, and completeness correlated strongly with overall quality ($R^2 \approx 0.9$) — **supporting checklist-coverage as the automated scoring proxy**. [9]

3.4 The perception gap (why the survey works) — now quantified

- ✓ 3–0 In the Handoff CEX nursing validation, handoff **participants rated quality 8.1 while trained external observers rated the same handoffs 7.1** — a 1.1-point inflation (95% CI 0.5–1.6, $p<0.001$). That is the quantified basis for the survey's predict-then-compare design: the student records their perception, the evaluator sets it against the measured coverage map. [5][6]
- ✓ 3–0 Structured handoff TRAINING demonstrably moves performance: an 18-hour SBAR shift-report program (83 staff nurses) took satisfactory handoff knowledge from 4.8% to 92.8% ($p<.001$; 74.7% retained at 3 months) and observed practice adequacy from 2.4% to 100% (90.4% at 3 months) — repeatable structured practice with observation works, and the platform automates exactly that expensive loop. [11]

3.5 High-risk element list (synthesis of the instruments above)

✓ 2–1 Direct literature support: under plain SBAR, **code status and pending examination results are frequently omitted** — confirming both as first-class high-risk elements here. Omissions flagged **HIGH-RISK** in the evaluator (the instructor's "would put a patient at risk" list): **allergies · code status · medication changes** (new/held/dose-changed/last-given/due-next, infusions running) · **pending results or tasks** · **lines/drains/access** · **active safety risks** (falls, airway, sepsis watch, bleeding precautions) · **anticipatory guidance / contingencies** · **abnormal vitals trend** · any **active staged discrepancy** (FR-008).

4 · The platform design

4.1 The handoff context pack — the ground truth

Generated per patient from the live session (chart seed + events + medication board + staged-error state) at handoff time. It is simultaneously: the AI counterpart's knowledge, the completeness rubric, and the debrief artifact.

#	Element	Source in the system	Risk
1	Identity & situation (name, age, room, admitting problem, today's story)	chart banner, chief complaint, encounter	
2	Illness severity, one line (stable / watcher / unstable)	vitals trend vs condition ranges	**HIGH**
3	Background: diagnoses, relevant history, **allergies**, **code status**	problem list, allergies, code_status	**HIGH**
4	Current assessment: latest vitals + trend direction, key labs, focused exam notes	vitals_baseline + vitals.record events, labs	
5	Medications & treatments: new / changed / held / due-soon, infusions running, high-alert flags	MAR + admin history, med board, IV fluids	**HIGH**
6	Lines / drains / access	chart (iv_fluids, devices)	**HIGH**
7	Pending items: orders, labs, consults awaiting	orders, events	**HIGH**
8	Safety risks (falls, airway, bleeding, isolation…)	condition + overlays + safety_class	**HIGH**
9	Anticipatory guidance: "watch for X — if X, then Y"	condition profile + active staged errors	**HIGH**
10	Receiver synthesis (read-back) occurred	transcript	**HIGH**
11	Responsibility transfer explicit ("I've got them")	transcript	

4.2 Two student roles, one machinery

	Student OFF-GOING (gives report)	Student ONCOMING (receives report)
AI counterpart	Oncoming nurse / charge nurse (from the existing character list) — holds the context pack, listens, then probes 2–4 omissions prioritizing high-risk ones, and asks for a synthesis if the student doesn't offer one	Off-going nurse — delivers report at an instructor-set **completeness dial**: complete · typical-gaps (the evidence-backed common omissions: anticipatory guidance + a pending item) · staged-error embedded (FR-008 report encounter)
Assessed behavior	Element coverage of what the student SAID + response quality to probes	The QUESTIONS the student asks (mapped to the gap list), read-back/synthesis, and whether planted gaps/discrepancies were caught

Containment	Same rule as the staged-error system: the AI introduces nothing beyond its instructions, never hints anything is staged, stays in character

4.3 Multi-patient mode (charge-nurse turnover)

Sequential context packs for 2–3 patients (v1 cap, ratify) + one cross-patient element: **prioritization** — "who do you see first and why." Builds on the shared-staff investigation (FR-007): one charge-nurse character handles all patients' reports in one conversation.

4.4 The post-handoff verbal survey

Presented on the station immediately after the handoff; the student answers each question by holding to talk; the room transcriber (already ratified, in-room only) writes the answers down. Questions (v1 set, ratify):

1. "Overall, how complete was the handoff you just gave/received, 0–10 — and why?" *(prediction)*
2. "Name the three most important things you handed off / heard." *(prioritization recall)*
3. "What, if anything, do you think you missed or wish you'd asked?" *(self-identified gaps)*
4. "What should be watched for on this patient over the next shift?" *(anticipatory guidance recall — the #1 evidence-backed omission)*
5. "Was there anything in the chart or report that didn't match what you heard or read?" *(discrepancy probe — fires the FR-008 comparison when an error was staged)*
6. *Oncoming only:* "What is your first action for this patient?" *(synthesis into plan)*

4.5 Evaluation & debrief

- **Coverage map:** each pack element scored binary (said / not said) against the handoff transcript — AI-assisted on the existing comparison-store pattern, every line instructor-confirmable before it is shown to the student. High-risk misses called out in red.
- **Perception delta:** the survey's self-rating and claimed coverage set against the measured map — "you rated yourself 8/10 and named meds and vitals; the record shows allergies, the pending culture, and the watch-for items were never mentioned."
- **Receiver metrics (oncoming):** questions asked vs the gap list; synthesis performed y/n; planted gap/discrepancy caught y/n.
- **Debrief artifact:** one section in the existing debrief surface — pack, transcript, coverage map, survey answers, perception delta, high-risk misses — printable for post-sim discussion. Formative, not a grade (recommended).

4.6 Composition (the instructor's "scenario built from scenarios")

Two entry points, same engine: **(a)** a handoff PHASE appended to any live session ("end the shift" button in the control room); **(b)** a STANDALONE handoff session built from the charts of one or more existing scenarios — pick patients on the Setup stage, the engine seeds their charts, the session IS the turnover.

5 · Implementation strategy

Stage	Scope	Builds on (already shipped)	Effort
H1	Context-pack generator + high-risk tagging + ground-truth element list	chart seed/fold, med board, FR-008 state	M
H2	Handoff session mode + AI counterpart prompts (receiver-probe / giver-with-dial, containment, synthesis ask, responsibility transfer)	FR-001/002/003 prompt machinery, persona list	M
H3	Multi-patient sequencing + prioritization element	H1–H2, FR-007 direction	M
H4	Verbal survey station flow (question sequence → PTT answers → storage)	FR-006b room transcription, audio station	M
H5	Evaluation engine (coverage map, survey comparison, perception delta, instructor confirm) + debrief section	comparison store, debrief builder	L
H6	Control-room integration (Setup: phase/role/dial/characters · Live: start-handoff + status) + field validation script	FR-005 two-stage control room	M

Sequencing: FR-008's remaining stages (S5 builder page, S6 debrief) come first — H5/H6 share those surfaces. Then H1→H6 in order; H1+H2 alone already yield a usable single-patient handoff drill.

6 · Risks & open questions

- **Scoring validity:** binary coverage measures completeness, not judgment quality — mitigated by instructor confirmation and formative use. *Ratify: formative-only for v1.*
- **AI receiver question quality:** the probing behavior needs prompt iteration in field tests (same path FR-001's decisive-doctor took).
- **Survey transcription accuracy:** drug names in answers ride the existing vocabulary-hints lever; the pack's terms join the hints during survey mode.
- **Session length (multi-patient):** cap at 2–3 patients for v1. *Ratify.*
- **Verification caveat** on the research basis (Section 3 box): key claims flagged for spot-check.

7 · References

[1] AHRQ Making Healthcare Safer IV — Structured Handoffs rapid review. effectivehealthcare.ahrq.gov/sites/default/files/related_files/structured-handoff-rapid-research.pdf

[2] BMJ Quality & Safety (2025) — MHS IV structured-handoff certainty ratings. pmc.ncbi.nlm.nih.gov/articles/PMC12232517/

[3] Journal of Advanced Nursing (2025) — I-PASS-structured bedside nursing handover pilot. onlinelibrary.wiley.com/doi/10.1111/jan.16896

[4] AHRQ TeamSTEPPS 3.0 — Handoff (information + authority + responsibility). ahrq.gov/teamstepps-program/curriculum/communication/tools/handoff.html

[5] Handoff CEX — nursing shift-report validation. pmc.ncbi.nlm.nih.gov/articles/PMC3504166/

[6] Handoff CEX — multi-site validation, rater effects, receiver-version limits. pmc.ncbi.nlm.nih.gov/articles/PMC3621018/

[7] SBAR-LA — 10-item binary learner-assessment rubric. ncbi.nlm.nih.gov/pmc/articles/PMC8520891/

[8] SBAR change-of-shift instrument — content validation + element checklist. ncbi.nlm.nih.gov/pmc/articles/PMC9749775/

[9] Simulated signout analysis (2024, n=177) — contingency omissions, receiver synthesis, completeness↔quality. ncbi.nlm.nih.gov/pmc/articles/PMC11430516/

[10] Penu et al. (2026) — scoping review: I-PASS implementation in nurse handovers. journalmpci.com/index.php/jhnr/article/view/1073 (cited with source-tier + evidence-table cautions; confirmed 3–0)

[11] SBAR shift-report training program, 83 staff nurses (pre/post + 3-month retention). pmc.ncbi.nlm.nih.gov/articles/PMC9969440/ (confirmed 3–0)

Additional consulted sources (13 more, incl. TeamSTEPPS curriculum pages, simulation-pedagogy and multi-patient handoff literature) are listed in the research run log.